Advanced Functions

Section 1: Exponential Functions & Growth/Decay

1 A bacterial culture doubles in population every 4 hours. If the initial count is 500, what is the population after 14 hours?

a. $4,000$

b. $4,000\sqrt{2}$

c. $8,000$

d. $16,000$

2 An investment of $2,000$ earns 6% annual interest compounded continuously. What is the value after t years?

a. $2000(1.06)^{t}$

b. $2000e^{0.06t}$

c. $2000(1+\frac{0.06}{12})^{12t}$

d. $2000e^{6t}$

3 Solve the exponential equation for x: $4^{x+1}=8^{2x-1}$.

a. $\frac{1}{2}$

b. $\frac{5}{4}$

c. $\frac{3}{2}$

d. 2

Section 2: Logarithmic Laws & Equations

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4 Solve for x: $\log_2(x) + \log_2(x-2) = 3$.

a. 4

b. 2

c. -2

d. 4 and -2

5 Which of the following is equivalent to $\frac{\ln 25}{\ln 5}$?

a. $\ln 5$

b. $\ln 20$

c. 2

d. 5

6 If $\log_b 3 = x$ and $\log_b 5 = y$, what is $\log_b(45)$ in terms of x and y?

a. $2x + y$

b. $x^2 + y$

c. $2xy$

d. $x + 2y$

Section 3: Rational Functions & Asymptotes

7 What is the slant (oblique) asymptote of the rational function $f(x) = \frac{2x^2 + 5x - 3}{x + 1}$?

a. $y = 2x + 3$

b. $y = 2x - 1$

c. $y = 2x + 5$

d. $y = 2x$

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8 Where does the rational function $g(x) = \frac{x^2 - 4}{x^2 - x - 2}$ have a removable discontinuity (hole)?

a. $x = 2$

b. $x = -1$

c. $x = -2$

d. No holes

9 What is the domain of $h(x) = \ln(x^2 - 9)$?

a. $(-3, 3)$

b. $(-\infty, -3) \cup (3, \infty)$

c. $[3, \infty)$

d. $x \neq \pm 3$

Section 4: Parametric & Polar Curves

10 Eliminate the parameter t in the parametric equations $x = 3\cos t$ and $y = 3\sin t$ to find the Cartesian equation.

a. $x^2 + y^2 = 9$

b. $x^2 - y^2 = 9$

c. $y = 3x$

d. $x + y = 3$

11 A curve is defined by $x(t) = 2t - 1$ and $y(t) = t^2 + 3$. What is y expressed directly as a function of x?

a. $y = \frac{(x+1)^2}{4} + 3$

b. $y = (2x - 1)^2 + 3$

c. $y = \frac{x^2 + 1}{2} + 3$

d. $y = \frac{(x-1)^2}{4} + 3$

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12 Convert the polar equation $r = 4\cos\theta$ into an equivalent Cartesian equation.

a. $(x-2)^2 + y^2 = 4$

b. $x^2 + (y-2)^2 = 4$

c. $x^2 + y^2 = 4$

d. $(x-4)^2 + y^2 = 16$

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